

1 Calcium modeling of spine apparatus-containing human
2 dendritic spines demonstrates “all-or-nothing” communication
3 switch between the spine head and dendrite

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Dendritic spines are highly dynamic neuronal compartments that control the synaptic transmission between neurons. Spines form ultrastructural units, coupling synaptic contact sites to the dendritic shaft and often harbor a spine apparatus organelle, composed of smooth endoplasmic reticulum, which is responsible for calcium sequestration and release into the spine head and neck. The spine apparatus has been linked recently to synaptic plasticity in adult human cortical neurons. While the morphological heterogeneity of spines and their intracellular organization has been extensively demonstrated in animal models, the influence of spine apparatus organelles on critical signaling pathways, such as calcium-mediated dynamics, is less well known in human dendritic spines. In this study we used serial transmission electron microscopy to carry out detailed anatomical reconstructions of nine human cortical spines in order to quantify and analyze the architectural plan of the spines and to incorporate the reconstructions into detailed modeling and simulation of the calcium dynamics between spine and dendrite. The anatomical study of reconstructed human dendritic spines revealed that the size of the postsynaptic density correlates with spine head volume and that the spine apparatus volume increases alongside the spine volume. Using a newly developed simulation pipeline, we have linked these findings to spine-to-dendrite calcium communication. While the absence of a spine apparatus, or the presence of a purely passive spine apparatus did not enable any of the reconstructed spines to relay a calcium signal to the dendritic shaft, the calcium-induced calcium release from this intracellular organelle allows for finely tuned “all-or-nothing” spine-to-dendrite calcium coupling; controlled by spine morphology, neck plasticity, and ryanodine receptors. Our results suggest a strategic position of spine apparatus organelles in the neck of human dendritic spines and their potential relevance in maintaining and regulating spine-to-dendrite calcium communication.

Keywords: calcium, dendritic spines, endoplasmic reticulum, intracellular calcium stores, Ryanodine receptors, human cortex

49 Introduction

50 Calcium-based second-messenger signaling pathways mediate fundamental biological
51 processes in the human body [1]. In the brain, synaptically evoked intracellular
52 Ca^{2+} surges regulate the ability of neurons to adjust their synaptic contacts in
53 an activity-dependent manner [2, 3]. This fundamental biological mechanism (i.e.,
54 synaptic plasticity), plays a crucial role in complex brain functions, such as internal
55 representation and memory formation [4, 5]. Over the past few decades, cellular and
56 molecular mechanisms of Ca^{2+} dependent synaptic plasticity have been extensively
57 studied in dendritic spines across various animal models [6, 7].

58 Dendritic spines are highly dynamic neuronal compartments on which cortical
59 excitatory synapses are typically formed [8, 9]. Most spines have a bulbous head and
60 a thin neck that connects the spine head to the shaft of the dendrite. Additionally,
61 they often harbor a dynamic extension of the smooth endoplasmic reticulum (sER)
62 in the dendrite, which can rapidly enter and leave the spine compartment and is
63 capable of changing its position inside the neck and head of spines [10]. Indeed,
64 they play a critical role in sER intracellular Ca^{2+} storage and in synaptic plasticity
65 [11, 12, 13, 14]. The variable nature of spine morphology and intracellular Ca^{2+} stores
66 has been studied with respect to electrical and biochemical signaling [15, 16, 17].
67 More recent research has studied the influence of synaptic spine architecture on
68 calcium signaling by parametrically designing the leading features of spines (i.e.,
69 spine neck width/length, spine head volume, and the size of the sER) and performing
70 computational studies on a wide variety of such synthetically designed geometries
71 [18, 19, 20].

72 Guided by the existing literature and our own recent work [20, 21, 22], we carried
73 out detailed anatomical reconstructions of human cortical spines based on serial
74 transmission electron microscopy and performed a correlation analysis on a set of
75 geometric features, including the size and position of the postsynaptic density (PSD),
76 spine head volumes, and cross-sectional areas of the spine neck. We then attempted
77 to answer how the distinct three-dimensional ultrastructural organization of human
78 dendritic spines influences intracellular Ca^{2+} dynamics in the spine head and neck,
79 and how these signals propagate into the attached dendritic shaft (i.e., spine-to-
80 dendrite Ca^{2+} communication). Specifically, we wanted to learn more about the role
81 of the spine apparatus organelle (SA) [23, 24, 25], an enigmatic cellular organelle
82 composed of stacked sER, found in the majority of cortical dendritic spines in humans
83 [22].

84 Results

85 Spine Ca^{2+} modeling in reconstructed human dendritic spines 86 containing a spine apparatus organelle

87 To capture the effect of the SA on Ca^{2+} dynamics in dendritic spines, the three-
88 dimensional architecture of dendritic spines, including intracellular membrane struc-
89 tures, must be carefully considered. Expanding on the results obtained from synthe-
90 tically generated dendritic spines [20], we reconstructed detailed morphologies of nine
91 dendritic spines, their SA, and postsynaptic densities from cortical surgical access
92 material using serial transmission electron microscopy (TEM). From the serial TEM
93 image stacks, we then generated realistic three-dimensional geometric models, see
94 Figures 1A and 1B.

95 Immunogold staining for the actin-binding protein synaptopodin, which is an
96 essential component for the formation of the SA in the rodent brain [26], confirmed
97 that the actin-modulating protein synaptopodin is a marker for human SA (Figure
98 1C; c.f., [22]). We then attached the reconstructed SAs to a single standardized
99 dendritic ER tubule in a dendritic compartment of identical dimensions for all spines
100 (Figure 1D). We used the resulting surface triangulation of the plasma membrane,
101 the SA, the PSD, and dendritic ER to construct a volumetric tetrahedral mesh [27]
102 for numerical simulations.

103 Next, we adapted and integrated existing single-channel models of Ca^{2+} exchang-
104 ers in the plasma membrane - i.e., plasma membrane Ca^{2+} -ATPase (PMCA) and
105 $\text{Na}^+/\text{Ca}^{2+}$ exchangers (NCX), as well as Ryanodine receptors (RyR) on the SA
106 membrane (see schematics in Figure 1E) - into a diffusion-reaction model for cy-
107 tosolitic and endoplasmic Ca^{2+} dynamics [20]. Numerical simulations were carried
108 out on computational domains defined by the reconstructed human dendritic spine
109 morphologies, using established numerical methods (details provided in Methods).

110 Spine apparatus-containing human dendritic spines showed 111 larger spine head volumes and postsynaptic densities

112 We recently showed that $\sim 70\%$ of all dendritic spines on human layer II/III pyrami-
113 dal neurons contain a synaptopodin cluster (i.e., SA) [22]. By carefully examining
114 EM cross-sections, we obtained evidence that large human dendritic spines contain
115 SA organelles [22], which is consistent with earlier work conducted on synaptopodin
116 clusters, SA, and spine head sizes of neurons in rodent brains [28, 12, 29, 30]. How-
117 ever, no three-dimensional analyses of SA-containing human dendritic spines have

118 been performed thus far.

119 To test whether there are correlations between the reconstructed spine compo-
120 nents, we carefully examined critical morphological parameters of the reconstructed
121 SA-containing dendritic spines (Figure 2). Consistent with previous studies in ani-
122 mal models [31, 32], a positive correlation between PSD sizes and spine volumes was
123 observed in the spines reconstructed from human cortical tissue (Figure 2B). This
124 relation was readily detected when spines were sorted by spine head volume. More-
125 over, we observed a positive correlation between PSD sizes and cross-sectional areas
126 of spine necks (Figure 2C). This finding raises the possibility that larger spines with
127 a wider neck may allow for a more effective Ca^{2+} spine-to-dendrite communication.

128 Following the major scope of this study (i.e., the modeling of spine-to-dendrite
129 Ca^{2+} communication), we determined distances between PSDs and the dendritic
130 compartment. Figure 2D shows no significant correlation between spine volumes and
131 PSD-to-dendrite distances, which were defined as the Euclidian distance between the
132 point of the PSD closest to the base of the spine. These results indicated that SA-
133 containing spines of the human cortex may differ in their volume and spine neck
134 diameters, while PSD-to-dendrite distances are comparable; at least, for the nine
135 reconstructed spines used in this study.

136 Finally, we examined the morphological properties of the SAs contained in the
137 reconstructed spines (Figure 2E). We observed a positive correlation between spine
138 volume and SA volume, which was also observed when spine neck cross-sections
139 were considered. These results provided ultrastructural morphological insight into
140 the three-dimensional organization of SA-containing dendritic spines in the human
141 cortex. They support the previously proposed spine-within-a-spine ER morphology,
142 which plays the leading role governing spine volume and sER volume ratio in Ca^{2+}
143 signaling [1, 18].

144 **Passive spine apparatus organelles have no major impact on** 145 **spine-to-dendrite Ca^{2+} signaling**

146 To assess the role of the SA in spine-to-dendrite Ca^{2+} communication, we first in-
147 vestigated Ca^{2+} signal propagation in spines for which the reconstructed SA was
148 excluded. In all simulations the same Ca^{2+} ion flux (j_c^{rls}) was used for the PSD
149 at the spine head, [33, 6], see Table 4 for the flux value. Changes in $[\text{Ca}^{2+}]$ were
150 measured in three regions of interest: the spine head, spine neck, and the dendrite
151 [20]. For all of our numerical experiments we simulated an AMPA model at the PSD
152 with a 10 ms release of Ca^{2+} by a linearly decreasing Ca^{2+} pulse, described further
153 in the *methods* section. The number of ions released at the PSD for each spine is

154 computed by

$$\# \text{ of ions} = \left[N_A \cdot \int_0^{\tau_{rls}} j_c^{rls} \cdot \left(1 - \frac{t}{\tau_{rls}} \right) \cdot A_{PSD} dt \right],$$

155 where A_{PSD} is the area of the PSD, and $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$ is Avogadro's
156 constant.

157 As shown in Figure 3A, only a small fraction of released Ca^{2+} ions reached the
158 dendritic compartment. To quantify the dendritic response to Ca^{2+} release at the
159 PSD, the $[\text{Ca}^{2+}]$ amplitude was measured in the spine head, neck and dendrite. The
160 values and ratios between head and dendrite amplitude for all spines are listed in
161 Table 1. In all cases less than 3% of the initial $[\text{Ca}^{2+}]$ amplitude in the head was
162 detectable in the dendrite. When the SA was added as a passive cellular compartment
163 (i.e., when it was present as a geometric obstacle, but without any Ca^{2+} exchange
164 mechanisms that would allow Ca^{2+} exchange across the SA membrane), similar and
165 near-zero dendritic $[\text{Ca}^{2+}]$ profiles were observed in the respective spines (Figure 3B).
166 Minor differences in the profiles (compared to the no-SA case) can be observed in
167 the head and neck; however, these all lie in the single-digit $\mu\text{mol/l}$ range.

168 The differences in peak $[\text{Ca}^{2+}]$ amplitude between passive-SA and no-SA are
169 presented in Figure 3C–E. These differences decreased with increasing spine volume
170 (Figure 3C) and are positively correlated to the SA/Spine volume ratio (Figure 3D).
171 Two clear outliers in this trend are spines 1 and 4. To test whether the SA is able to
172 block Ca^{2+} passage at the intersection between head and neck [34], and thus produce
173 larger differences in the $[\text{Ca}^{2+}]$ amplitude, the data was plotted against the SA/spine
174 surface area ratio at this intersection (Figure 3E). The outlier, spine 4, could clearly
175 be explained by this metric. Thus, accumulation of SA in the intersectional region
176 between head and neck may obstruct Ca^{2+} diffusion. Numerical simulations were
177 also carried out for PSD adjusted Ca^{2+} influx (see Figure S1)- i.e. Ca^{2+} influx was
178 scaled by PSD surface area, using spine 8 as the reference spine. It should be noted
179 that PSD-adjusted Ca^{2+} influx may compensate for the Ca^{2+} accumulation in the
180 head region. In Figure S1, for example, the change in $[\text{Ca}^{2+}]$ remains relatively
181 uniform when correlated against spine volume, with the exception of spine 4 for
182 which we observe the above -described “corking effect”. Taken together, from these
183 results (fixed Ca^{2+} and PSD-adjusted), we can conclude that spine-to-dendrite Ca^{2+}
184 signaling in the dendrite is negligible for both the no-SA and the passive-SA setting.
185 These observations strengthen the case for active Ca^{2+} exchange across the spine ER
186 membrane in order to enable spine-to-dendrite Ca^{2+} communication in SA-containing
187 dendritic spines.

188 Ryanodine receptors of spine apparatus control spine-to-dendrite 189 Ca^{2+} signaling

190 Previous work revealed that RyR-dependent Ca^{2+} -induced Ca^{2+} release from synapto-
191 podin-associated Ca^{2+} stores modulates spine Ca^{2+} dynamics and synaptic plasticity
192 [12, 14, 7]. To assess the role of active SAs in human dendritic spines, by including
193 RyRs on the membrane of the SA and dendritic ER, we determined the critical RyR
194 density for spine-to-dendrite Ca^{2+} communication for each spine (Table 2). To quan-
195 tify spine-to-dendrite Ca^{2+} coupling, we defined the peak $[\text{Ca}^{2+}]$ to be a concentration
196 value greater than $1 \mu\text{mol}/l$ in the dendritic measuring zone for 1 ms. Converting
197 this concentration minimum (during 1 ms) leads to $1 \times 10^{-18} \text{ mol}/(s \cdot \mu\text{m}^3)$, and
198 then multiplying by the width of the dendrite ($\sim 1 \mu\text{m}$) obtains an approximate flux
199 through the neck-dendritic interface of $1 \times 10^{-18} \text{ mol}/(s \cdot \mu\text{m}^2)$. Therefore, we de-
200 fined a spine-to-dendrite coupling as successful if more than 11% of the initial Ca^{2+}
201 influx at the PSD was measured in the dendrite (using $8.6 \times 10^{-18} \text{ mol}/(s \cdot \mu\text{m}^2)$ for
202 reference). We increased the RyR density in $0.01 \mu\text{m}^{-2}$ increments in the range of
203 0–5 RyRs μm^{-2} and found critical RyR values for all reconstructed morphologies,
204 ranging between 1.37 and $3.5 \mu\text{m}^{-2}$ (Table 2). Figure 4A shows all spines eliciting
205 dendritic Ca^{2+} coupling when the critical RyR density is surpassed. A small number
206 of RyRs on SAs (i.e., 1–6 receptors) was sufficient to enable spine-to-dendrite Ca^{2+}
207 communication, see Table 2. However, the peak $[\text{Ca}^{2+}]$ drops below $1 \mu\text{mol}/l$ when
208 the RyR density is subcritical (Figure 4B). Thus, RyR density plays a critical role
209 in spine-to-dendrite Ca^{2+} coupling.

210 We next tested whether changes in Ca^{2+} entry at the PSD affect the critical
211 RyR density by successively increasing the Ca^{2+} flux density at the PSD entry site.
212 However, coupling occurs at the same RyR density, independent of the initial Ca^{2+}
213 influx within the range $8.6 \times 10^{-18} - 22.0 \times 10^{-18} \text{ mol}/(s \cdot \mu\text{m}^2)$ tested here, see Figures
214 4C-D for spine 8 as an example. This invariance can be attributed to intracellular
215 Ca^{2+} buffering, as modifications in the buffering capacity produced a shift in the
216 critical RyR density. This was observed when we ran our simulations with half
217 the buffering capacity (Figure 4E), which caused a small decrease in the critical
218 RyR density. It is also worth clarifying that the transition between coupling and
219 decoupling spine-to-dendrite Ca^{2+} signals is highly nonlinear (see jumps in Figures
220 4C-E). Changes in critical RyR density appear to control the “all-or-nothing” Ca^{2+}
221 dynamics in the parent dendrite.

222 Comparing the peak calcium concentrations of passive and active SA in the head
223 and dendrites (see Table 1) showed that the relative differences between passive and
224 active ER dynamics are the result of Ca^{2+} -induced Ca^{2+} release through RyRs, which

releases additional Ca^{2+} into the cytosol [35]. While the critical RyR density was not dependent on Ca^{2+} influx at the PSD, we further analyzed whether the structural organization of dendritic spines could play a role. For completeness, we provided Ca^{2+} profiles (see Figure S2) for PSD-adjusted Ca^{2+} influx at both critical RyR and subcritical RyR densities.

Spine morphology and neck plasticity affect spine-to-dendrite Ca^{2+} signaling

Critical for endoplasmic Ca^{2+} release through RyRs is having sufficient $[\text{Ca}^{2+}]$ at the endoplasmic membrane [36, 37, 14]. Thus, an intracellular organization of the SA that would support sufficient accumulation of Ca^{2+} at the spine neck (the most confined part of the spine morphology) should reduce the critical RyR density. To test this hypothesis, the critical RyR density was plotted against the ratio of SA in the spine neck to the spine neck cross-sectional area (Figure 5A). We found that the area ratio correlated negatively with the RyR density, demonstrating that a smaller RyR density is sufficient to induce spine-to-dendrite coupling when the SA occupies larger portions of the spine neck.

Interestingly, we observed a positive correlation between the spine and SA cross-sectional area of the spine neck (Figure 5B), suggesting that the SA adjusts to dendritic spine growth and reorganizes in such a way to support spine-to-dendrite Ca^{2+} communication. Previous work has demonstrated that changes in spine neck diameters accompany the induction of excitatory synaptic plasticity, reflected by an increase in spine head size [38, 39, 40]. Specifically, spine necks are shortened and widened after long-term potentiation of excitatory neurotransmission [39].

We therefore systematically tested whether changes in spine neck diameter can affect spine-to-dendrite Ca^{2+} communication. To address the role of spine neck plasticity in spine-to-dendrite Ca^{2+} communication, we ran simulations by using the reconstruction of spine 6, and manually widening the spine neck by 10%, 20%, and 40%, while keeping the RyR density and the SA morphology constant (Figure 5, modified spines 6-A, 6-B, 6-C, respectively). While the Ca^{2+} dynamics in the spine head remained unchanged (Figure 5C), we observed that increasing the spine neck width caused a decrease in peak Ca^{2+} near the SA membrane, thereby producing insufficient Ca^{2+} release from the SA store to maintain spine-to-dendrite coupling (Figure 5D).

We then tested whether an increase in RyR density could balance morphological plasticity in order to maintain coupling even under spine growth and reorganization. For each of the modified spines 6-A, 6-B, and 6-C, we incrementally increased the

261 RyR density and observed that a critical RyR density can be determined for all
 262 spines. Naturally, this critical value increases with increasing neck width, see Figure
 263 5E (critical RyR densities for spines 6-A, 6-B, 6-C can be found in Table 2). We have
 264 not tested whether increasing the size of the SA restores spine-to-dendrite commu-
 265 nication. The current results appear to suggest that for a SA with subcritical RyR
 266 density, increasing the SA size will restore spine-to-dendrite communication.

267 This means, that not only could coupling be restored but, as Figure 5F shows,
 268 the dendritic Ca^{2+} profiles for all adjusted spines are nearly identical, indicating that
 269 Ca^{2+} is sensitive to changes in spine neck and SA morphology in human dendritic
 270 spines, and that Ca^{2+} dynamics are affected by spine neck and SA geometry [41, 42,
 271 43].

272 In conclusion, we observed an “all-or-nothing” Ca^{2+} communication switch be-
 273 tween spine head and dendrite that is controlled by an interplay between RyR density
 274 and morphological changes of dendritic spines and SAs.

275 Discussion

276 In this study, we explored spine Ca^{2+} dynamics in human cortical spines using sim-
 277 ulations consisting of (i) Ca^{2+} diffusion and reaction with mobile calcium buffer
 278 (calbindin, CalB) in the cytosol; (ii) Ca^{2+} diffusion inside the SA and sER of the
 279 dendrite; and (iii) Ca^{2+} exchange across the plasma (PMCA, NCX) and endoplasmic
 280 (RyR) membranes (Figure 1E). We showed that a critical RyR density, dependent
 281 on the overall spine and SA architecture, allows active Ca^{2+} -induced Ca^{2+} release
 282 to maintain spine-to-dendrite Ca^{2+} communication. Increasing the neck diameter or
 283 decreasing the SA cross-sectional area in the neck region of individual spines disrupts
 284 this communication pathway. Increasing the RyR density (or SA sizes) can reverse
 285 this effect (Figure 6). These findings suggest a critical role for SA- and spine neck
 286 plasticity in controlling Ca^{2+} communication between the spine head and dendrite.

287 The ability to visualize individual dendritic spines in living brain tissue over
 288 extended periods was developed three decades ago [44, 45]. Consequently, the amount
 289 of information about the morphology and physiology of dendritic spines and the
 290 molecular mechanisms acting on spines has increased tremendously [41, 46, 47, 4, 48].
 291 However, major issues regarding structure-function interrelations and the role of
 292 dendritic spines in network function and complex behavior remain unresolved [48].
 293 This is particularly true for structure-function interrelations in the human cortex,
 294 which are beginning to emerge [22]. In adult human neurons, it is currently not
 295 possible to carry out the required experiments to study spine Ca^{2+} dynamics as
 296 they require genetic tools for simultaneous visualization of dendritic spines, SAs and

changes in intracellular Ca^{2+} levels. Also, it is impossible to systematically assess the relevance of individual spine and SA/ER parameters, as they are not easy to manipulate in biologically complex systems. In this respect, computer simulations based on serial EM reconstructions provide suitable tools to explore spine Ca^{2+} dynamics in human dendritic spines.

Our morphological analyses of the nine reconstructed spines used in this study confirm and expand on our recent findings on dendritic spine plasticity of layer II/III pyramidal neurons in the adult human cortex [22]. In this earlier study, we showed that (i) approximately 70% of dendritic spines contain synaptopodin clusters; (ii) synaptopodin clusters and spine apparatus organelles are found in dendritic spines with large cross-sectional areas; and (iii) a plasticity-inducing stimulus promotes remodeling of synaptopodin clusters, spine apparatus organelles, and dendritic spines in human cortical neurons. These changes correlated well with changes in spontaneous excitatory postsynaptic currents recorded from individual layer II/III pyramidal neurons [22]. Consistent with these earlier findings, we confirmed in this study that synaptopodin is a marker for the human SA, and we also demonstrated a positive correlation between SA volumes, PSD sizes, and spine volumes. Although we concede that this analysis is currently based only on nine reconstructed spines, disclosing such interrelation with comparably small numbers indicates that these morphological measures may be tightly regulated in the human cortex.

Our simulations using a purely passive SA, strengthen the case for active Ca^{2+} exchange across the spine ER membrane to enable spine-to-dendrite Ca^{2+} communication in SA-containing dendritic spines. Specifically, choosing a critical RyR density enabled spine-to-dendrite Ca^{2+} communication. This RyR density depended on the overall spine and SA architecture, while being independent of the strength of the initial Ca^{2+} influx through the PSD. By increasing spine neck sizes for individual spines this communication pathway was disrupted. Increasing the RyR density, or adjusting SA to spine-neck ratios, however, reversed this effect. While the actual RyR densities and changes in the number of RyRs on ER membranes remain unknown, changes in spine neck sizes and spine apparatuses (both sizes and position within a spine) are well described [12, 29, 39, 10, 30]. Hence, the physiology and morphology of dendritic spines appear tightly integrated and can control Ca^{2+} communication between the spine head and dendrite. In this context, the SA may serve Ca^{2+} homeostasis in maintaining spine-to-dendrite Ca^{2+} communication during dendritic spine growth as seen after plasticity induction. Conversely, it is interesting to theorize that changes in SA sizes or position within the spine may allow for rapid input-specific control of spine Ca^{2+} signals. Whether such mechanisms serve input-specific synaptic plasticity, Ca^{2+} dependent local dendritic protein synthesis [49], synaptic

335 tagging [50, 51], or other as yet unknown mechanisms warrants further investigation.
336 It may be important to mention in this context, however, that in our recent study we
337 showed that blocking protein synthesis with anisomycin halted plasticity induction,
338 dendritic spine, and SA growth in adult human cortical neurons [22], suggesting a
339 role of local protein synthesis in SA-dependent synaptic plasticity.

340 Interestingly, the transition between coupling and decoupling of spine-to-dendrite
341 Ca^{2+} signals was highly nonlinear. This “all-or-nothing” property observed in our
342 simulations is reminiscent of the physiological function of another cellular compart-
343 ment in which synaptopodin-associated stacked ER is present: the axon initial seg-
344 ment and the cisternal organelle [52, 53, 54, 55, 56, 57]. Like SAs, cisternal organelles
345 consist of flat, membrane-delineated cisternae alternating with electron-dense ma-
346 terial. Similarly, synaptopodin-deficient mice do not form SAs and cisternal or-
347 ganelles [55]. Indeed, from an anatomical point of view the analogy between spine
348 heads and spine necks, and the soma and axon initial segment is striking. How-
349 ever, while the role of SA in spine calcium dynamics and synaptic plasticity seems
350 well-established, the biological significance of cisternal organelles and their role in
351 AIS-plasticity [58, 59, 60] remains less understood. It is probable that increased use
352 of new electron microscopy techniques on human cortical tissue, such as focused ion
353 beam, block-face serial electron microscopy, and array tomography, together with
354 computer-based automatic 3D reconstructions will allow us to advance toward a
355 better understanding of ultrastructure-function relations of synaptopodin associated
356 stacked smooth ER in distinct neuronal compartments, and to test the extent to
357 which data obtained from animal models translates to the adult human brain.

358 **Methods**

359 **Ethics statement**

360 Human brain tissue was obtained from a local biobank operated in the Department
361 for Neurosurgery at the Faculty of Medicine, University of Freiburg Germany (AZ
362 472/15_160880). All procedures were carried out after positive evaluation by the
363 local ethics committee (AZ 593/19).

364 **Electron microscopy**

365 Tissue was fixed in 4% paraformaldehyde (w/v) and 2% glutaraldehyde (w/v; phos-
366 phate buffered saline) overnight. After fixation tissue was cut into 100 μm thin
367 slices, thoroughly washed in phosphate buffer (0.1M) and incubated with 1% osmium

368 tetroxide for 60 minutes. After washing in graded ethanol (up to 50% /v/v) for 10
369 minutes each, slices were incubated with uranyl acetate (1% (w/v) in 70% (v/v)
370 ethanol) overnight. Slices were then dehydrated in graded ethanol (80%, 90%, 96%
371 for 10 minutes each, and 2x 100% for 10 minutes). After two washing steps in propy-
372 lene oxide for 5 minutes each, the slices were incubated with durcupan/propylenoxid
373 (1:1 for 1 hour) following durcupan only overnight at room temperature. Ultra-
374 thin sections of 20 - 40 nm were prepared at a Leica UC6 Ultracut. Sections were
375 mounted into copper grids (Plano) and contrasted using Pb-citrate for 3 minutes.
376 Transmission electron microscopy was performed at a Zeiss Leo 906 equipped at
377 6000x magnification. In total, cortical access tissue from 7 individuals who under-
378 went clinically indicated neurosurgical procedures, e.g., for tumors or epilepsy, were
379 processed and 1 to 6 acquisitions were obtained from each patient. Each acquisition
380 included at least one labeled dendritic segment and synaptic contact. Up to 40 im-
381 ages of the same dendritic spines on consecutive sections were acquired and saved as
382 TIF-files.

383 **3D-reconstruction of dendritic spines and generation** 384 **of spine models**

385 Image series were adjusted in brightness and contrast using the Fiji software pack-
386 age [61]. Stack alignment and segmentation were performed with the integrated
387 TrakEM2 plugin [62]. For this purpose, an automatic alignment (rigid registration)
388 was utilized followed by manual correction. Independent traces were drawn manu-
389 ally for each spine attached to a dendritic segment, spine apparatus organelles and
390 postsynaptic densities. Dendritic spines possessing spine apparatus organelles that
391 were complete within the series were reconstructed. A total of 9 series met these
392 criteria and were used in analysis. The image segmentation binary files were trans-
393 formed into a preliminary surface mesh using a Java 3D library [63] and saved in an
394 ‘OBJ’ format. However, meshes generated in this manner contained various mesh
395 artifacts, such as jagged boundaries, non-manifold features and surface intersections.
396 Using the meshing software ProMesh [64] for post-processing, including Laplacian
397 smoothing, optimized surface meshes, compatible with physical simulations, were
398 produced. Mesh defects, such as disconnects in the spine apparatus organelle due to
399 thickness in ultrathin section or errors in the segmentation, were manually identified
400 and matched to the cellular structures. Spine structures were manually subdivided
401 into head and neck region after visual inspection of the 3D model from different
402 angles. All reconstructed spines, including their spine apparatus organelles, were cut
403 from their parent dendritic shaft through the base of the spine neck. Subsequently,

they were attached to an dendritic segment containing the endoplasmatic reticulum. Using this method, we inserted spine apparatuses into reconstructed human spine models and generated hybrids of anatomical morphologies. Using ProMesh, we measured volumes from the three-dimensional reconstructions. While spine neck length was measured from the center of the spine head base towards the insertion of the spine into the dendrite, neck diameter was calculated as an average diameter from three measurements obtained proximal, intermediate and distal to the edge of the dendrite. Linear regressions were performed by best-fit approaches and were statistically tested to be different from zero with the statistical software GraphPad Prism (GraphPad Software).

Calcium Simulations

All modeling components were implemented in the simulation toolbox NeuroBox [65]. NeuroBox is a simulation toolbox that combines models of electrical and biochemical signaling on one- to three-dimensional computational domains. NeuroBox allows the definition of model equations, typically formulated as ordinary and partial differential equations, of the cellular computational domain and specification of the mathematical discretization methods and solvers [66, 67]. The microscopy images were processed to generate 3D surface mesh geometries. With these surface mesh geometries we used Promesh [64] to perform a tetrahedralization of the surface geometries [27, 68], in particular we performed a Delaunay tetrahedrization which avoids tetrahedra with particularly acute or obtuse interior angles.

Model Equations

Spatio-temporal calcium dynamics are modeled by a system of diffusion-reaction equations

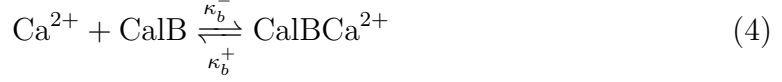
$$\frac{\partial c_c}{\partial t} = D_c \Delta c_c + (\kappa_b^- (b^{tot} - b) - \kappa_b^+ b c_c), \quad (1)$$

$$\frac{\partial b}{\partial t} = D_b \Delta b + (\kappa_b^- (b^{tot} - b) - \kappa_b^+ b c_c), \quad (2)$$

$$\frac{\partial c_e}{\partial t} = D_c \Delta c_e, \quad (3)$$

where (c_c) denotes $[Ca^{2+}]$ in the cytosol, (c_e) the concentration in the ER, and b the calbindin- D_{28k} concentration. The diffusion constants D are defined using data from [69, 70]. The interaction between cytosolic calcium and calbindin- D_{28k} (CalB) are

described by



where the concentration of the CalB- Ca^{2+} compound is expressed by the difference of the total concentration of CalB present in the cytosol (b^{tot}) and free CalB, the former of which is assumed to be constant in space and time (this amounts to the assumption that free calcium and CalB have the same diffusive properties). All constants and parameters are borrowed from [20] but are provided in Table 4 for completion. While CalB has four distinct high-affinity Ca^{2+} -binding sites [71], we treat it as though it had only one, at the same time quadrupling its concentration in our model. This amounts to assuming that all four binding sites are essentially equal and binding is non-cooperative. The rate constants κ_b^+ and κ_b^- are defined in [20].

Membrane transport mechanisms

In order to study the influence of the intracellular organization on Ca^{2+} signals, we included Ca^{2+} exchange mechanisms on the plasma membrane (PM). For the plasma membrane, we considered plasma membrane Ca^{2+} -ATPase pumps (PMCA), $\text{Na}^+/\text{Ca}^{2+}$ exchangers (NCX), calcium release due to voltage-dependent calcium channels (VDCC), and a leakage term. This amounts to the flux equations (number of ions per membrane area and time)

$$j_{ERM} = j_R + j_{l,e}, \quad (5)$$

$$j_{PM} = -j_P - j_N + j_{l,p} \quad (6)$$

where j_R is the RyR flux density and $j_{l,e}$ is the leakage flux density on the ERM, and j_P , j_N and $j_{l,p}$ the flux densities of PMCA, NCX, and leakage flux density of the PM, respectively. Homogeneous distributions of all exchange mechanisms were assumed, as experimental data on precise numbers and spatio-temporal distribution of these receptors within individual spines are not available.

RyR Channels

The Ca^{2+} flux density generated by ryanodine receptor channels at the ER membrane is given by an expression of the form

$$j_R = \rho_R \cdot p_R^o \cdot I_R, \quad (7)$$

where ρ_R is the density of RyR in the ER membrane, p_R^o is the open state probability of a single channel, and I_R is the single channel Ca^{2+} current. Using [72] we defined a single channel as

$$I_R = I_R^{ref} \frac{c_e - c_c}{c_e^{ref}}, \quad (8)$$

where the reference current I_R^{ref} is approximated in [73]. The open probability of the RyR channels is borrowed from [72] and is computed by the sum of two open probabilities o_1 and o_2 which are governed by the system

$$o_1 = 1 - c_1 - o_2 - c_2 \quad (9)$$

$$\frac{\partial c_1}{\partial t} = k_a^- o_1 - k_a^+ c_c^4 c_1 \quad (10)$$

$$\frac{\partial o_2}{\partial t} = k_b^+ c_c^3 o_1 - k_b^- o_2 \quad (11)$$

$$\frac{\partial c_2}{\partial t} = k_c^+ o_1 - k_c^- c_2 \quad (12)$$

where the kinetic constants $k_a^\pm, k_b^\pm, k_c^\pm$ are given in Table 4 and this system was solved independently on every point of the ER surface geometry.

PMCA Pumps

We used the model given in [74, 75], where the plasma membrane Ca^{2+} -ATPase current is modeled as a second-order Hill equation

$$j_P = \rho_P \cdot \frac{I_P c_c^2}{K_P^2 + c_c^2}. \quad (13)$$

NCX Pumps

For the $\text{Na}^+/\text{Ca}^{2+}$ exchange current we assumed a constant Na^+ concentration at the plasma membrane utilizing a first-order Hill equation from [74, 75]:

$$j_N = \rho_N \cdot \frac{I_N c_c}{K_N + c_c}. \quad (14)$$

Leakage Term

The ER membrane and plasma membrane have a leakage flux that needs to be accounted for and incorporated with described transport mechanisms. The leakage

474 fluxes are calculated to ensure there is zero membrane net flux in the equilibrium
 475 state for all simulated ions and agents. Leakage flux densities were modeled by

$$\begin{aligned} j_{l,e} &= v_{l,e} \cdot (c_e - c_c), \\ j_{l,p} &= v_{l,p} \cdot (c_o - c_c), \end{aligned}$$

476 where c_o is the extracellular Ca^{2+} concentration, which is assumed constant.

477 Calcium Influx

478 The Ca^{2+} influx is modeled using Neumann boundary conditions for the cytosolic
 479 Ca^{2+} concentration, i.e. the time-dependent influx density function is defined at the
 480 postsynaptic portion of the plasma membrane, an AMPA model. For our experiments
 481 we modeled release of Ca^{2+} by a linearly decreasing Ca^{2+} pulse for $\tau_{rls} = 10$
 482 millisecond duration starting at an initial maximal specific current density j_c^{rls}

$$j_c^{rls} \cdot \left(1 - \frac{t}{\tau_{rls}}\right). \quad (15)$$

483 We assumed a constant membrane potential $V_m = -70$ mV.

484 Numerical Methods

485 For numerical simulations, the four equations were discretized in space using a finite
 486 volumes method. Current densities, both synaptic and across the ER and plasma
 487 membranes, were incorporated into the reaction-diffusion process very naturally and
 488 easily this way. We show how this is achieved using the cytosolic Ca^{2+} equation as
 489 an example: It is reformulated (using the divergence theorem) to an integral version

$$\int_B \frac{\partial c_c}{\partial t} dV = \int_{\partial B} D_c \nabla^T c_c \cdot n_{\partial B} dS + \int_B (\kappa_b^- (b^{tot} - b) - \kappa_b^+ b c_c) dV \quad (16)$$

490 where B is a control volume that will be specified shortly, and $n_{\partial B}$ is the outward
 491 normal on the boundary of B . For control volumes located at the ER membrane,
 492 some portion of its boundary will coincide with the ER membrane. Since there is no
 493 diffusive flux density $D_c \nabla c_c$ across the ER membrane, we could simply substitute
 494 it by the ER flux density j_{ERM} into the boundary integral for this portion of the
 495 boundary. The same applies to the plasma membrane and the synapse area. The
 496 diffusive flux is set to zero on the rest of the cytosolic domain boundary. If we denote

the cytosolic boundary by Γ , its ER/plasma membrane and synaptic parts by Γ_{ERM} , Γ_{PM} and Γ_{syn} , respectively, this yields the following equation:

$$\begin{aligned} \int_B \frac{\partial c_c}{\partial t} dV &= \int_{\partial B \setminus \Gamma} D_c \nabla^T c_c \cdot n_{\partial B} dS + \int_{\partial B \cap \Gamma_{ERM}} j_{ERM}^T \cdot n_{\partial B} dS \\ &+ \int_{\partial B \cap \Gamma_{PM}} j_{PM}^T \cdot n_{\partial B} dS + \int_{\partial B \cap \Gamma_{syn}} j_{syn}^T \cdot n_{\partial B} dS \\ &+ \int_B (\kappa_b^- (b^{tot} - b) - \kappa_b^+ b c_c) dV \end{aligned} \quad (17)$$

Control volumes were constructed as a Voronoi-like dual tessellation of the original tetrahedral mesh by connecting the mid-points of edges, faces and volumes through planar facets. Equation (17) must hold for all control volumes, giving rise to one equation per control volume. Time discretization was realized using a backwards Euler scheme, i.e., for each point in time t , the term $\frac{\partial c_c}{\partial t}$ in (17) is replaced by the discretized term $\frac{c_c(t) - c_c(t - \tau)}{\tau}$ and all quantities on the right-hand side were evaluated at time t . Here, τ is the time-step size of the time discretization. By limiting the function space to the space of continuous functions that are linear on all volumes of the original mesh, the integrals in equation (17) can be evaluated efficiently. Moreover, the solution can be represented by one degree of freedom per volume, so there is one equation for each degree of freedom. The system of equations arising from this procedure is nonlinear (due to the nonlinear reaction term and, more importantly, the highly nonlinear transport terms across the membranes) and is therefore linearized by a Newton iteration. For the results we present here, the emerging linearized problems were solved using a Bi-CGSTAB [76] linear solver using Gauss-Seidel preconditioning. Computations were facilitated by a domain decomposition parallelization approach and carried out using the UG4 framework [66].

Compute Resources

This research includes calculations carried out on HPC resources supported in part by the National Science Foundation through major research instrumentation grant number 1625061 and by the US Army Research Laboratory under contract number W911NF-16-2-0189. The numerical simulations for this work were also carried out using the Extreme Science and Engineering Discovery Environment (XSEDE) see [77], which is supported by National Science Foundation grant number ACI-1548562. Through XSEDE we used the SDSC Comet HPC resources, allocation identification DMS200031.

525 **Acknowledgements**

526 We thank Barbara Joch for excellent technical assistance in electron microscopy. This
527 work was supported by NIH (R01MH118930 to GQ; 1R01NS109498 to AV) and by
528 the Federal Ministry of Education and Research, Germany (BMBF, 01GQ1804A to
529 AV)

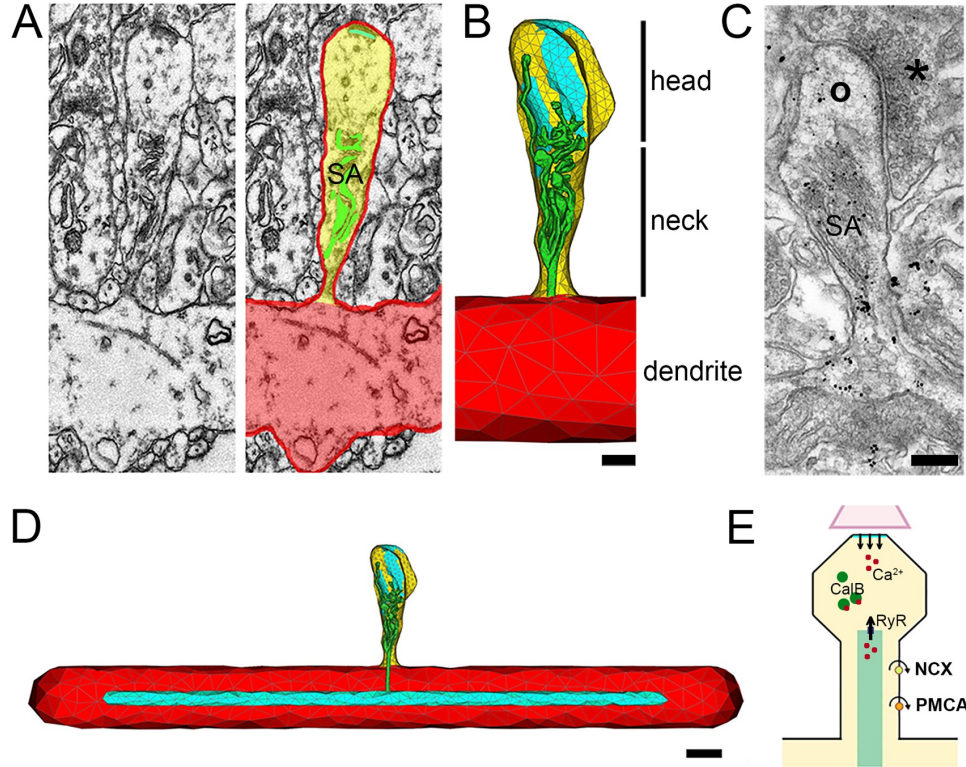


Figure 1: Dendritic spine reconstruction and Ca^{2+} modeling. (A) Spine morphologies are based on serial reconstructions of transmission electron micrographs obtained from human cortical tissue. (B) An example of a reconstructed dendritic spine (dendrite, red; spine compartment, yellow) carrying a spine apparatus organelle (SA, green). Upon release of Ca^{2+} at the reconstructed postsynaptic density (cyan), changes in $[\text{Ca}^{2+}]$ are determined in the head, neck and dendritic region, respectively. Scale bar, 200 nm. (C) Immunogold staining for the actin-binding protein synaptopodin [22], which is a marker and essential component of the SA. Asterisk indicates presynaptic, circle the postsynaptic compartment. Scale bar, 200 nm. (D) In all reconstructed human spines the SA was attached to a single standardized dendritic ER tubule in a dendritic compartment with identical dimensions. Scale bar, 500 nm. (E) The model accounts for Ca^{2+} exchange mechanisms on the plasma membrane ($\text{Na}^+/\text{Ca}^{2+}$ exchangers (NCX), plasma membrane Ca^{2+} -ATPases (PMCA)) and on the ER (green; ryanodine-receptors (RyR)). Calcium buffering capacity (Calbindin, CalB) is kept constant. Further details on physiological values are provided in the main text and in Table 4. Tables 2 and 3 contain morphological measurements such as surface area. See also supplemental movie.

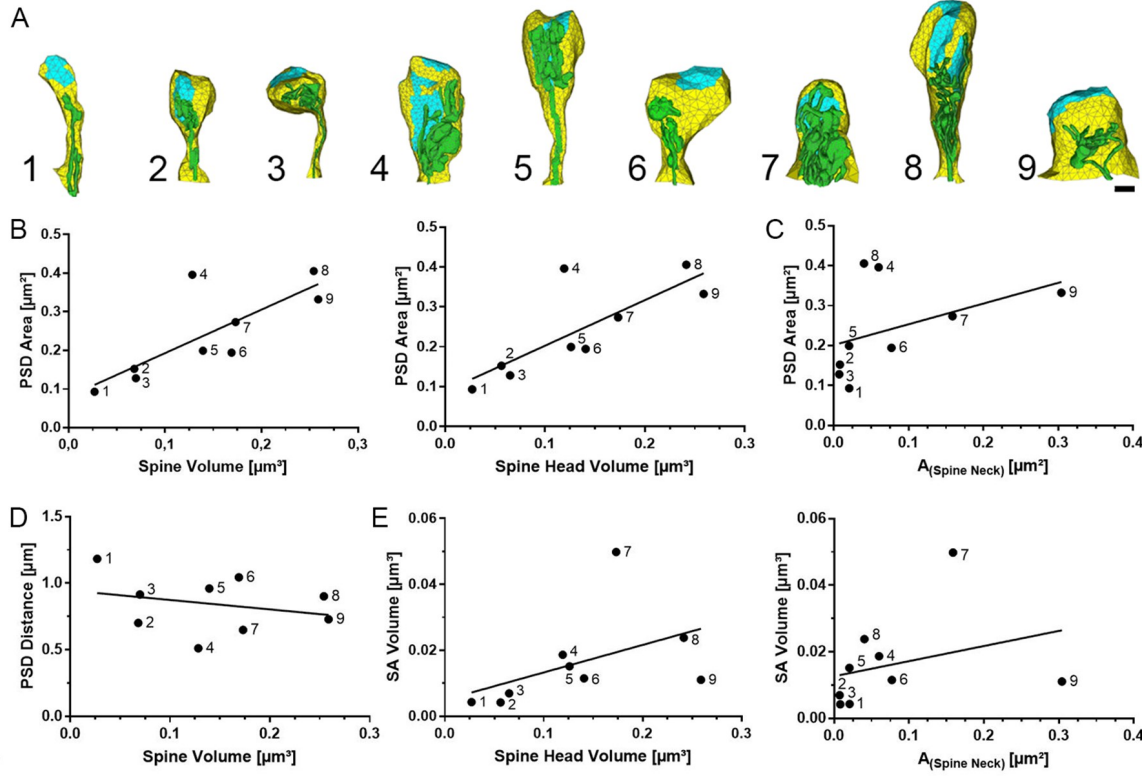


Figure 2: Morphological parameter of reconstructed spines. (A) Three-dimensional models based on reconstructed human dendritic spines. Spine apparatus organelle, green. Postsynaptic density, cyan. Scale bar, 200 nm (B) Dendritic spines with large volumes have large postsynaptic densities. (C) Positive correlation between spine neck diameter and postsynaptic density sizes. (D) Postsynaptic densities in large spines are not further away from spine base. (E) Large spines contain large spine apparatus organelles, which are found in spines with wider spine necks. $n = 9$ reconstructed dendritic spines from 4 independent samples.

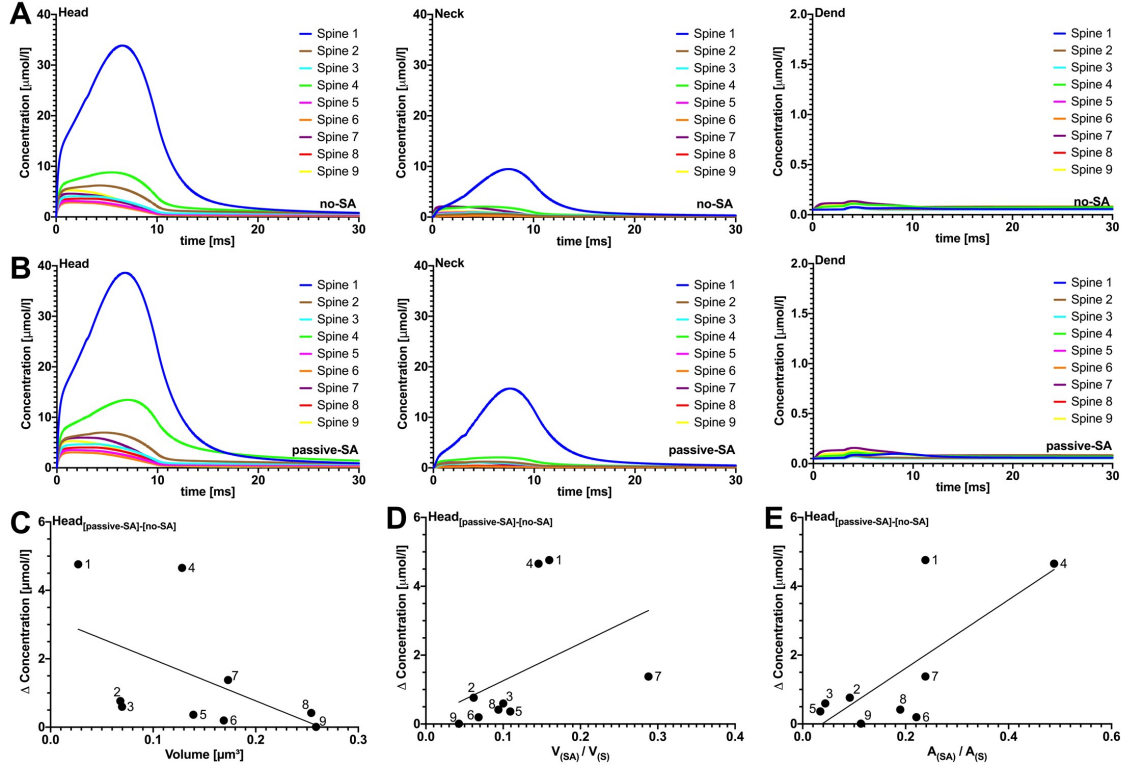


Figure 3: Effects of passive spine SA on spine-to-dendrite Ca^{2+} signaling, with fixed Ca^{2+} influx. Row A, without SA, and row B, with SA, show Ca^{2+} profiles for 10 ms initial Ca^{2+} release into the spine head. Consistent with experimental data, spine-to-dendrite Ca^{2+} signaling does not occur for all 9 simulated spines, and this also holds true for simulations where Ca^{2+} influx is adapted to the postsynaptic area, see additional plots. Row C (fixed Ca^{2+} influx) demonstrates correlations between maximum $[\text{Ca}^{2+}]$ in the head region versus spine volume, ratio of SA area/Spine area, and SA volume/Spine volume. Presence of a passive SA, i.e., no Ryanodine receptors, has no major effect on $[\text{Ca}^{2+}]$ dynamics in the spine head and neck.

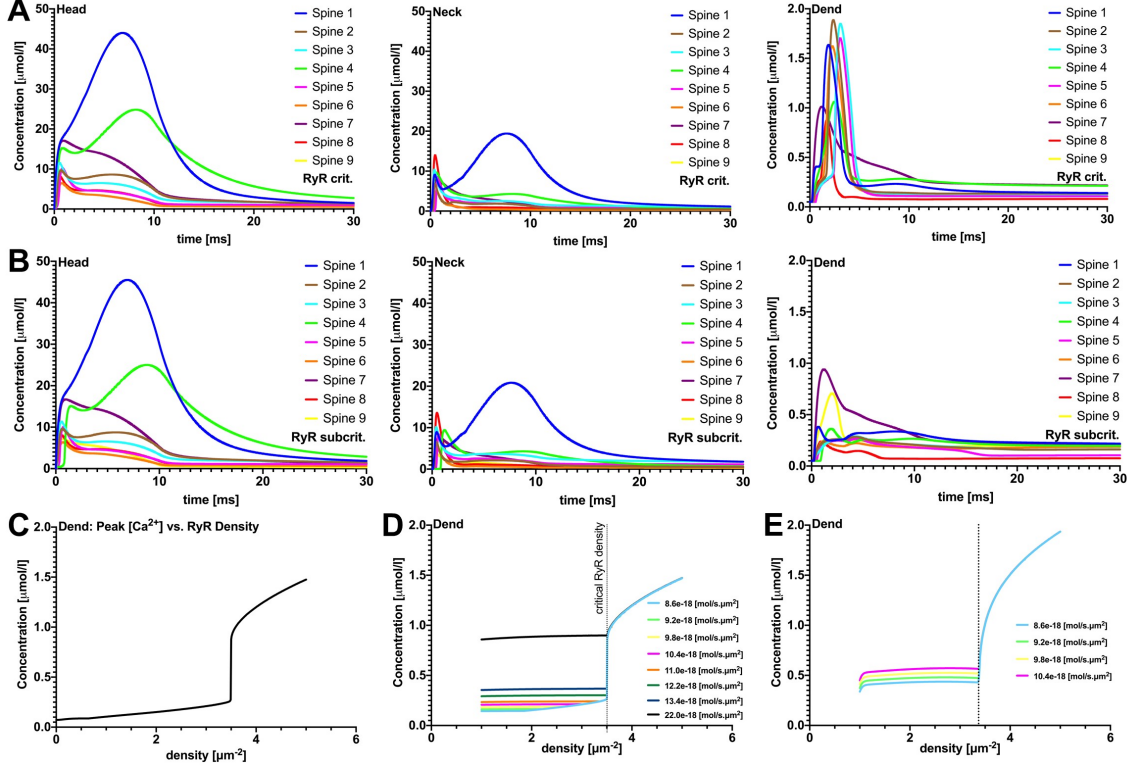


Figure 4: Effects of active SA with RyR calcium exchange mechanisms only. These plots were generated with active SA with only ryanodine receptors. Plots in row A show the calcium concentration traces in the head and dendrite at the critical RyR density. Plots in row B show the calcium concentration traces at sub-critical (5% below critical) RyR density. It is worth observing that Ca^{2+} coupling is significantly decreased. In Figures 4C-4E, we demonstrate for spine 8 that there is a critical RyR density such that Ca^{2+} coupling occurs in the dendritic region. The RyR density parameter is incremented in $0.01 \mu\text{m}^{-2}$ increments. One can observe the maximum $[\text{Ca}^{2+}]$ in the dendritic region has a significant jump at $\approx 3.50 \mu\text{m}^{-2}$. For plot D we performed the same RyR density increment experiments but with different Ca^{2+} influx at the synapse. For all spine simulations (rows A and B) we used a fixed Ca^{2+} influx at the synapse.

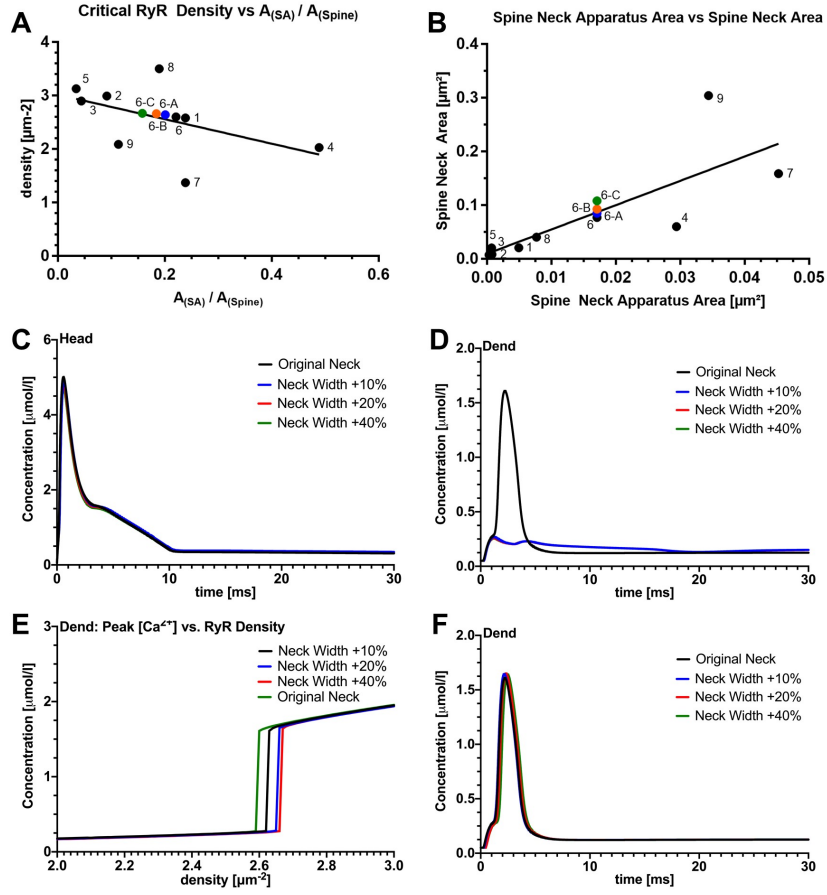


Figure 5: In A we plot the critical RyR density against the ratio of Spine Apparatus Neck area to Spine Neck area. In Table 2 we show the values that correspond to A. In C and D we demonstrate that widening the neck diameter of a spine affects the coupling of Ca^{2+} to the dendritic region. In particular E indicates that neck widening causes the critical RyR density to increase.

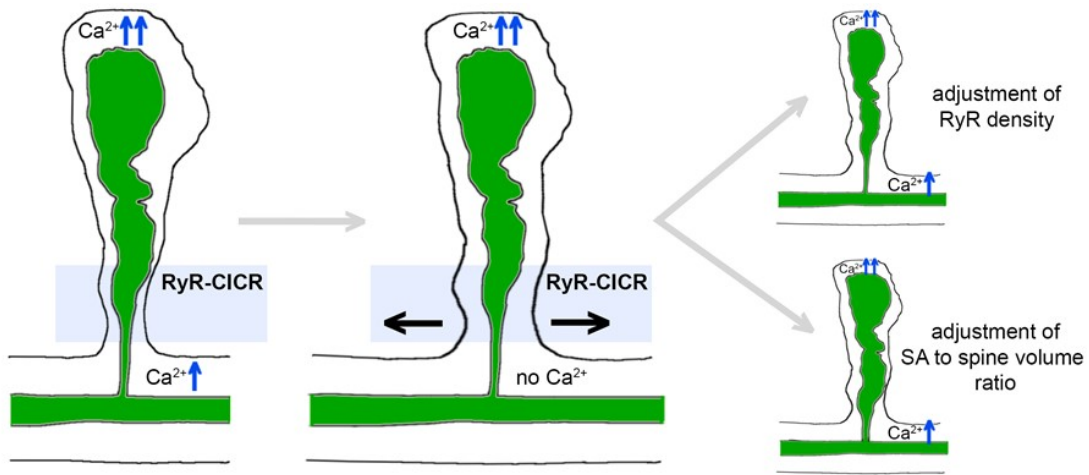


Figure 6: “All or-nothing” spine-to-dendrite calcium communication. Schematic illustration summarizing our major findings. Ca^{2+} communication between spine head and dendrite is controlled by the interplay between RyR density, and morphological changes of dendritic spines and spine apparatus organelles (SA) in the neck region. RyR, Ryanodine receptors.

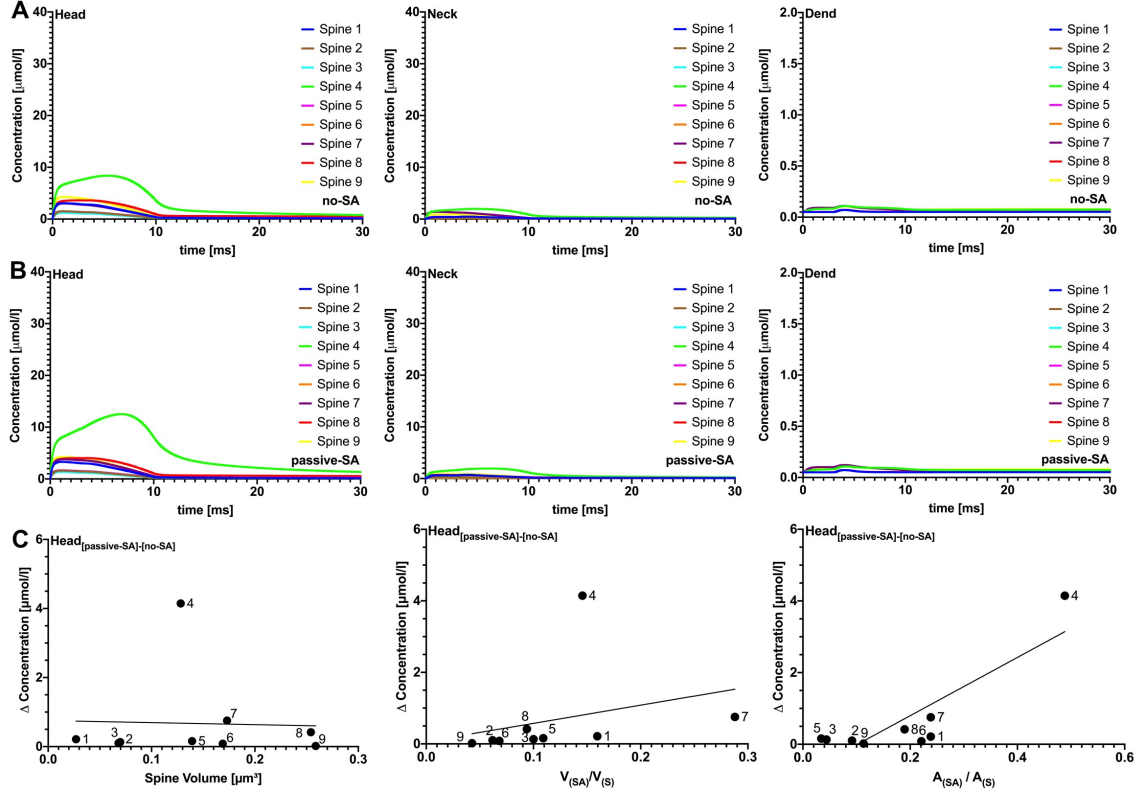


Figure S1: Effects of passive spine SA on spine-to-dendrite Ca^{2+} signaling, with PSD adjusted Ca^{2+} influx. Plots in row A are without SA and plots in row B are with SA, showing Ca^{2+} profiles for 10 ms initial Ca^{2+} release into the spine head. Consistent with experimental data, spine-to-dendrite Ca^{2+} signaling does not occur for all 9 simulated spines. Presence of a passive SA, i.e., no Ryanodine receptors, has no major effect on Ca^{2+} dynamics in the spine head and neck.

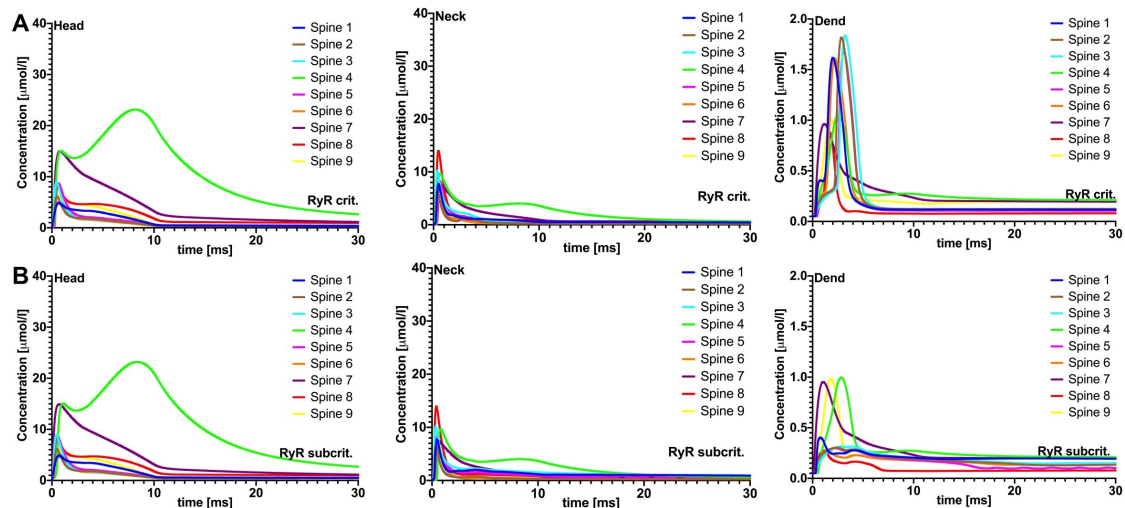


Figure S2: All plots were generated with PSD adjusted calcium influx. In row A, these are the calcium concentration profiles in the head and dendrite region at critical RyR density. In row B, these calcium concentration profiles correspond to sub-critical RyR density.

Spine No.	Head			Dendrite		
	passive peak [Ca ²⁺]	active peak [Ca ²⁺]	% increase	passive peak [Ca ²⁺]	active peak [Ca ²⁺]	% increase
1	33.85	44.00	30%	0.08	1.64	1,950%
2	6.22	9.66	55%	0.07	1.89	2,600%
3	4.09	11.47	180%	0.07	1.85	2,543%
4	8.80	24.82	180%	0.11	1.06	864%
5	3.16	10.22	223%	0.07	1.70	2,329%
6	2.86	6.47	126%	0.07	1.62	2,214%
7	4.60	17.09	270%	0.14	1.01	621%
8	3.62	8.11	124%	0.07	0.88	115%
9	5.29	7.20	36%	0.12	1.05	775%

Table 1: All concentrations are measured in $\mu\text{mol}/l$.

Spine No.	RyR Critical Density μm^{-2}	SA Surface Area	SA + Tube Surface Area	No. RyRs (SA only)	No. RyRs (SA + Tube)
1	2.58	0.3775	5.2696	1	14
2	2.99	0.3607	5.6055	1	17
3	2.90	0.5867	5.8341	2	17
4	2.03	1.1270	6.3761	2	13
5	3.13	1.0291	6.2729	3	20
6	2.60	0.6807	5.9312	2	15
6A	2.64	0.7092	5.9596	2	16
6B	2.66	0.7220	5.9724	2	16
6C	2.67	0.7347	5.9852	2	16
7	1.37	2.3209	7.5629	3	10
8	3.50	1.7146	6.6048	6	23
9	2.09	0.9134	5.8054	2	12

Table 2: Surface area is in units of μm^2 .

Spine No.	Area SA Neck	Area Spine Neck	Cross Section Ratio [%]
3	0.0003265	0.0074247	4.4
5	0.0007031	0.0205083	3.4
2	0.0007682	0.0083852	9.2
1	0.0049353	0.0206937	23.8
8	0.0076741	0.0404725	19.0
6	0.0170800	0.0772860	22.1
4	0.0294116	0.0602114	48.8
9	0.0343718	0.3041610	11.3
7	0.0452300	0.1587200	28.5

Table 3: Surface area is in units of μm^2 .

Initial and equilibrium values		PMCA pumps	
c_c	50 nM	I_p	$1.7 \times 10^{-23} \text{ mol} \cdot \text{s}^{-1}$
c_e	250 μM	K_p	60 nM
c_o	2 mM	ρ_p	500 μm^{-2}
p	40 nM	NCX pumps	
b^{tot}	40 μM	I_N	$2.5 \times 10^{-21} \text{ mol} \cdot \text{s}^{-1}$
Diffusion/reaction		K_N	1.8 μM
D_c	220 $\mu\text{m}^2 \cdot \text{s}^{-1}$	ρ_N	15 μm^{-2}
D_p	280 $\mu\text{m}^2 \cdot \text{s}^{-1}$	Leakage	
D_b	20 $\mu\text{m}^2 \cdot \text{s}^{-1}$	$v_{l,e}$	38 $\text{nm} \cdot \text{s}^{-1}$
κ_b^-	19 s^{-1}	$v_{l,p}$	4.5 $\text{nm} \cdot \text{s}^{-1}$
κ_b^+	27 $\mu\text{M}^{-1} \text{s}^{-1}$	Calcium release	
κ_p	0.11 s^{-1}	j_c^{rls}	$8.6 \times 10^{-18} \text{ mol} \cdot \text{s}^{-1} \mu\text{m}^{-2}$
p^r	40 nM	τ_{rls}	10 ms
RyR channel			
k_a^-	28.8 s^{-1}		
k_a^+	1500 $\mu\text{M}^{-4} \cdot \text{s}^{-1}$		
k_b^-	385.9 s^{-1}		
k_b^+	1500 $\mu\text{M}^{-3} \cdot \text{s}^{-1}$		
k_c^-	0.1 s^{-1}		
k_c^+	1.75 s^{-1}		
ρ_r	0.0 – 5.0 μm^{-2}		
I_R^{ref}	$3.5 \times 10^{-18} \text{ mol} \cdot \text{s}^{-1}$		

Table 4: Constants used in simulations are borrowed from [20].

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